Middle East Technical University
Department of Electrical and Electronics Engineering

# EE201 – Circuit Theory I

## Lecture 6: Dependent Sources and the Ideal Transformer

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| Colour code used in these notes and in the slides | | | |
|---|---|---|---|
| current $i$ | voltage $v$ | power $p$ | magnetic flux $\Phi$ |

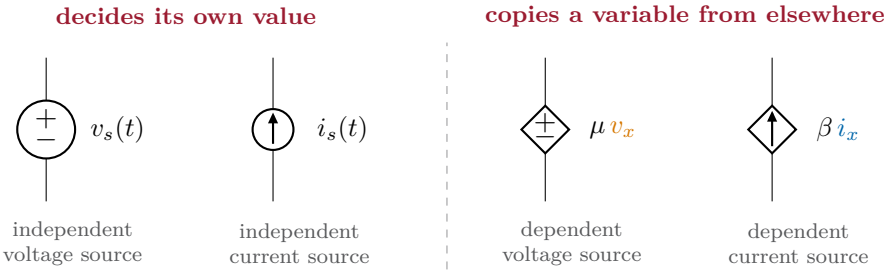
## 1. Sources controlled from elsewhere

The sources of Lecture 3 decide their own value. A 12 V source holds 12 V whatever the rest of the circuit does. Today's sources do not: their value is set by a voltage or a current *somewhere else* in the circuit. They are how an amplifier, a transistor or an op-amp enters a circuit diagram.

**Dependent (controlled) source**

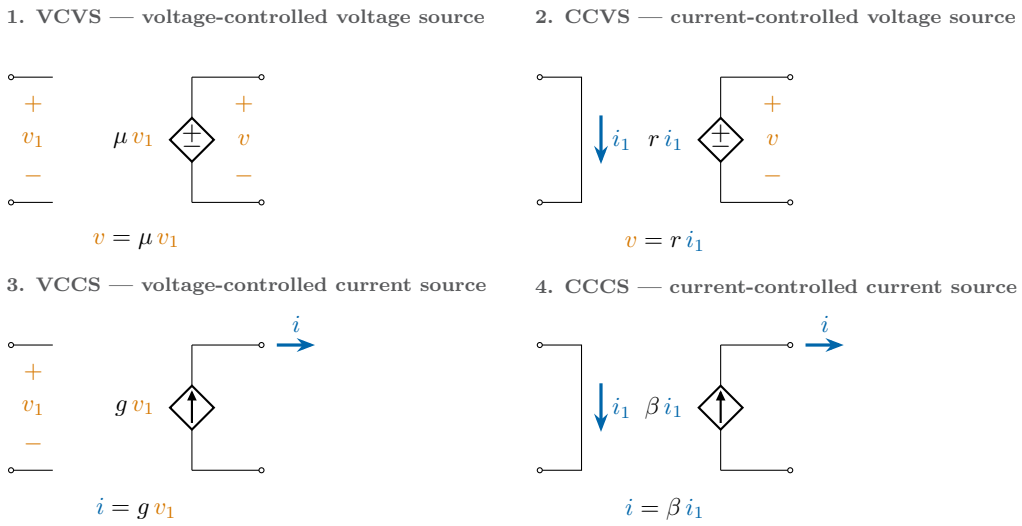
A **dependent source** is a voltage or current source whose value is a function of a voltage or current in *another* branch of the circuit, the **controlling variable**. When the function is a constant times the controlling variable, it is a **linear** dependent source. All dependent sources in EE201 are linear.

The symbol is a **diamond**. A circle means the source decides its own value; a diamond means it copies, scaled, a variable from elsewhere.




## 2. The four kinds

A dependent source senses a voltage or a current, and produces a voltage or a current. That gives four kinds. The name reads in that order — what it senses, then what it produces.



| Kind | Terminal equation | The constant is called | Unit |
|---|---|---|---|
| VCVS | $v = \mu\, v_1$ | voltage gain | none (V/V) |
| CCVS | $v = r\, i_1$ | transresistance | ohm, Ω (V/A) |
| VCCS | $i = g\, v_1$ | transconductance | siemens, S (A/V) |
| CCCS | $i = \beta\, i_1$ | current gain | none (A/A) |

The unit of the constant is always "what comes out" over "what goes in". When the two are different kinds of quantity the name carries *trans-*: a transresistance turns a current into a voltage, so it is measured in ohms, like a resistance.

### 2.1. The sensing side

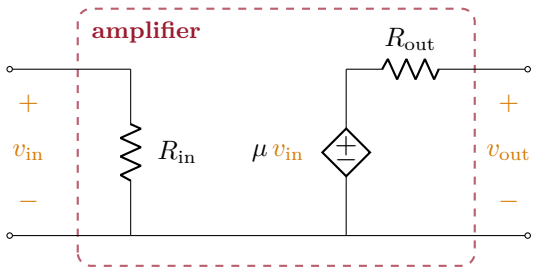
In each picture the left-hand pair of terminals is where the source senses its controlling variable. It senses it without disturbing anything:

- to sense a voltage it is an **open circuit** — it draws no current;
- to sense a current it is a **short circuit** — it takes no voltage.

In a real circuit diagram the sensing side is almost never drawn. The controlling variable is simply the voltage across, or the current through, some element that is already there, and the diagram labels it: $v_x$, $i_x$, $i_1$.

## 3. Why we need them

A dependent source is the simplest model of anything that *amplifies*. The picture below is how an amplifier usually appears in a circuit diagram.



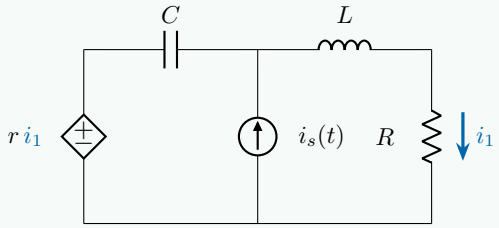

The input draws a small current through $R_{\text{in}}$. The output acts like a source $\mu\, v_{\text{in}}$ behind $R_{\text{out}}$. The gain $\mu$ is the only place where the output can grow larger than the input.

The same idea models real devices. A transistor is modelled with a VCCS or a CCCS; you will meet both in the electronics courses. An op-amp is a VCVS with an enormous gain, $\mu \approx 10^5$; it has a chapter of its own later in this course.

## 4. Reading a diagram with a dependent source

**Example 6.1 — what the diamond says**

Which equation does the dependent source contribute in this circuit?




It is a CCVS, so its terminal equation is $v = r\, i_1$ — and $i_1$ is the current through $R$. The source sits at the far left; the current it senses flows at the far right.

Count as in Lecture 2: $n = 4$, $b = 5$, so $2b = 10$ unknowns and $b = 5$ terminal equations. Four of them — for $C$, $L$, $i_s$ and $R$ — each speak about one element. The fifth ties the source's own voltage to another branch's current.

**What is new**

In Lecture 2 every terminal equation described one element *in isolation*. A dependent source does not: its terminal equation names a variable of another branch. Nothing else changes. KCL and KVL are the same, and there are still $2b$ equations for $2b$ unknowns.

**Two meanings of "controlled"**

In Lecture 3 a resistor was *voltage-controlled* if its current is a function of its *own* voltage, $i = f(v)$. A voltage-controlled *source* is different: its value is set by the voltage of *another* branch. The first describes one element's curve; the second is a link between two branches.

## 5. Solving circuits with dependent sources

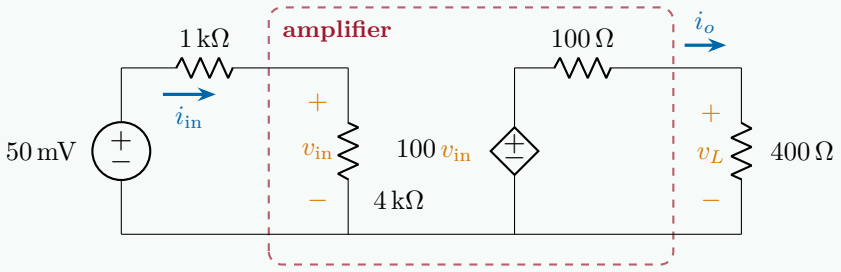
No new method is needed.

**Recipe**

1. Treat the dependent source as if it were independent, with its value written as $\mu v_x$, $r i_x$, $g v_x$ or $\beta i_x$.
2. Write KCL, KVL and the terminal equations as usual.
3. Add one equation that gives the controlling variable in terms of your unknowns.

**Example 6.2 — an amplifier**

A microphone, $v_s = 50\,\text{mV}$ with internal resistance $R_s = 1\,\text{k}\Omega$, drives the amplifier of Section 3 with $R_{\text{in}} = 4\,\text{k}\Omega$, $\mu = 100$ and $R_{\text{out}} = 100\,\Omega$. The load is $R_L = 400\,\Omega$. Find $v_L$, and compare the power the microphone supplies with the power the load receives.




**Input side.** One loop, a divider: $i_{\text{in}} = 50\,\text{mV}/5\,\text{k}\Omega = 10\,\mu\text{A}$, so $v_{\text{in}} = 4\,\text{k}\Omega \times 10\,\mu\text{A} = 40\,\text{mV}$.

**Output side.** The diamond now has a value: $100 \times 40\,\text{mV} = 4\,\text{V}$. Another divider: $i_o = 4\,\text{V}/500\,\Omega = 8\,\text{mA}$, so

$$\boxed{v_L = 400\,\Omega \times 8\,\text{mA} = 3.2\,\text{V}}, \qquad \frac{v_L}{v_s} = 64.$$

The overall gain is 64, not 100: part of the signal is lost in $R_s$ and part in $R_{\text{out}}$.

## Example 6.2, continued — where the power comes from

With the PSC on every element:

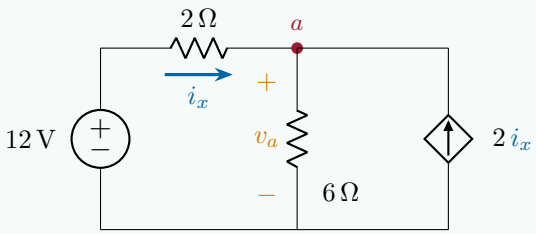
| input side | | output side | |
|---|---|---|---|
| microphone $v_s$ | $-0.5\,\mu$W | dependent source | $-32$ mW |
| $R_s$ | $+0.1\,\mu$W | $R_{\text{out}}$ | $+6.4$ mW |
| $R_{\text{in}}$ | $+0.4\,\mu$W | $R_L$ | $+25.6$ mW |

Each side sums to zero. The load receives 25.6 mW while the microphone supplies only 0.5 $\mu$W — 51 200 times less. The dependent source supplied the difference. A real amplifier takes that energy from its power supply, which the model does not draw.

One more detail: the bottom wire is shared, yet it carries no current. Wrap a closed surface around the input side (Lecture 2): that wire is the only branch crossing it, so its current is zero. The two sides interact only through $v_{\text{in}}$.

## Example 6.3 — when the controlling variable is unknown

The CCCS pushes $2i_x$ into node $a$, where $i_x$ is the current through the 2 Ω resistor. Find $i_x$ and $v_a$.




**Equations.** Here step 3 is free: $i_x$ is already one of the unknowns.

$$\text{KVL:}\quad -12+2i_x+v_a=0, \qquad \text{KCL at } a\text{:}\quad -i_x-2i_x+\frac{v_a}{6}=0 \implies v_a=18\,i_x.$$

**Solution.** $-12+2i_x+18i_x=0$, so

$$\boxed{i_x=0.6\text{ A}}, \qquad \boxed{v_a=10.8\text{ V}}.$$

**Power check.** The 12 V source supplies 7.2 W. The resistors absorb $2(0.6)^2=0.72$ W and $10.8^2/6=19.44$ W. The CCCS carries 1.2 A *out of* its + terminal, so it absorbs $10.8\times(-1.2)=-12.96$ W:

$$-7.2+0.72+19.44-12.96=0.$$

The 6 Ω resistor absorbs more than the independent source supplies. The dependent source makes up the rest.

## What a dependent source cannot do

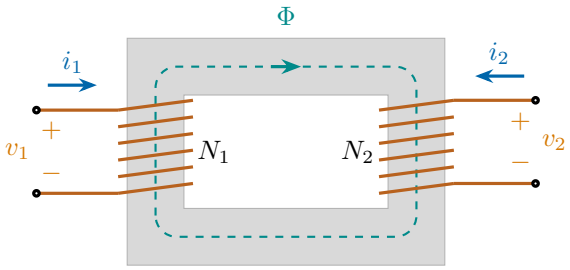
**It cannot start anything.** Switch the 12 V source of Example 6.3 off. The equations become $2i_x+v_a=0$ and $v_a=18i_x$, so $20\,i_x=0$: every voltage and current is zero, the dependent source's included. Without an independent source, all-zero satisfies every equation — and when the circuit has a unique solution, that is the solution.

**It is never switched off.** Later we will switch sources off one at a time (superposition, Thévenin equivalents). Only *independent* sources are switched off. A dependent source stays: its value is not an input, it is part of how the circuit behaves.

# 6. The ideal transformer

## 6.1. Two coils, one core

Lecture 1 assumed that no changing magnetic field links the loops of a circuit. Lecture 2 added: when we *want* such a link, we draw it as an element of its own. The transformer is that element. In Lecture 5 a coil's changing current made a changing field through the coil itself. Wind two coils on one iron core and the field of each passes through the other.




iron core: one flux Φ through both windings

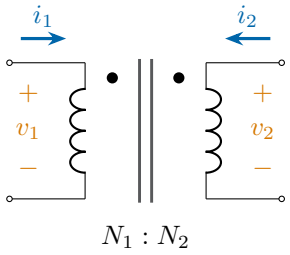
## 6.2. The element

**Ideal transformer (two windings)**

With $v_1$, $v_2$ positive at the dotted terminals and $i_1$, $i_2$ entering the dotted terminals,

$$\boxed{\frac{v_1}{N_1} = \frac{v_2}{N_2}} \qquad \boxed{N_1 i_1 + N_2 i_2 = 0}$$

$N_1$ and $N_2$ are the numbers of turns. Equivalently, $v_2 = \dfrac{N_2}{N_1}\, v_1$ and $i_2 = -\dfrac{N_1}{N_2}\, i_1$.




The two lines are the core. The **dots** mark the ends of the windings that go positive together; they act like reference marks. Move one dot to the other end of its winding and both equations change sign: $v_2 = -\frac{N_2}{N_1} v_1$ and $i_2 = +\frac{N_1}{N_2} i_1$.

## 6.3. Where the two equations come from

**Voltages.** One flux Φ passes through every turn of both windings. Each turn therefore has the same voltage, $d\Phi/dt$, and $N$ turns in series add up to $v = N\, d\Phi/dt$. Divide by $N$:

$$\frac{v_1}{N_1} = \frac{v_2}{N_2} = \frac{d\Phi}{dt}.$$

**Currents.** An *ideal* core needs no current at all to carry its flux. So the windings' currents, each weighted by its number of turns, must cancel: $N_1 i_1 + N_2 i_2 = 0$. A real core needs a small magnetising current; the ideal one needs none.

## 6.4. Any number of windings

One core may carry $k$ windings. The same two arguments give one ratio for all of them and one sum.

**Ideal transformer ($k$ windings)**

$$\frac{v_1}{N_1} = \frac{v_2}{N_2} = \cdots = \frac{v_k}{N_k} \qquad \text{and} \qquad N_1 i_1 + N_2 i_2 + \cdots + N_k i_k = 0,$$

every $v_j$ positive and every $i_j$ entering at the dotted terminal of winding $j$.

**No derivatives — and no DC**

The ideal transformer's equations are algebraic: no $d/dt$, no memory. It is a *resistive* element, listed with resistors and dependent sources in the course outline, even though it is built from coils.

A real transformer is not like that. It works only while the flux is changing. Feed it DC and, once the current has settled, the flux is constant: the secondary voltage is zero and the primary is just a coil — a short circuit at DC, as in Lecture 5. Use the ideal model for changing waveforms only.

## 7. The power of an ideal transformer

Each winding has its reference arrow entering its + terminal, so each obeys the PSC and $v_j i_j$ is the power that winding absorbs.

**The ideal transformer absorbs no power**

At every instant,

$$p(t) \;=\; \sum_{j=1}^{k} v_j i_j \;=\; 0.$$

**Proof.** Call the common ratio $\lambda(t) = v_j/N_j$, the same for every $j$. Then $v_j = N_j\lambda$, and

$$\sum_{j=1}^{k} v_j i_j \;=\; \sum_{j=1}^{k} N_j \lambda\, i_j \;=\; \lambda \sum_{j=1}^{k} N_j i_j \;=\; \lambda \cdot 0 \;=\; 0. \qquad \blacksquare$$

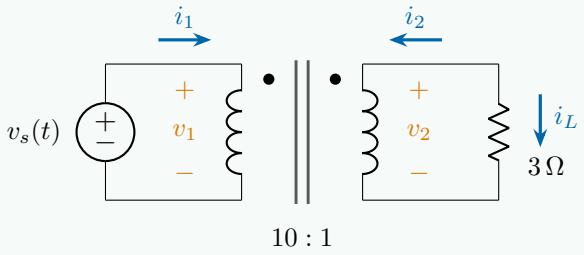
The first equation gave the common $\lambda$; the second made the sum vanish. Both are needed.

| Element | Absorbed power | What happens to the energy |
|---|---|---|
| resistor | $p \geq 0$ | dissipated |
| capacitor, inductor | $p$ can change sign | stored, then returned |
| ideal transformer | $p = 0$ at every instant | passed straight through |

With two windings, $v_1 i_1 = -v_2 i_2$: whatever enters the primary leaves the secondary at the same instant. The voltage is multiplied by $N_2/N_1$ and the current divided by the same factor, so the product is unchanged.

**Example 6.4 — a step-down transformer**

A 10 : 1 transformer connects $v_s(t) = 120\cos\omega t\,\mathrm{V}$ to a $3\,\Omega$ load. Find the currents and the power on each side.




**Voltages.** $v_1 = 120\cos\omega t\,\mathrm{V}$, so $v_2 = \frac{1}{10}v_1 = 12\cos\omega t\,\mathrm{V}$.

**Currents.** The load takes $i_L = v_2/3 = 4\cos\omega t\,\mathrm{A}$. That current leaves the dotted terminal, so $i_2 = -4\cos\omega t\,\mathrm{A}$, and $i_1 = -\frac{N_2}{N_1}i_2 = 0.4\cos\omega t\,\mathrm{A}$.

**Power.**

$$p_1 = v_1 i_1 = 48\cos^2\omega t\,\mathrm{W}, \qquad p_2 = v_2 i_2 = -48\cos^2\omega t\,\mathrm{W}, \qquad p_1 + p_2 = 0.$$

The primary takes $48\cos^2\omega t\,\mathrm{W}$ from the source and the secondary hands exactly that to the load — at every instant, not only on average.

**Seen from the source.** $v_1/i_1 = 120/0.4 = 300\,\Omega$. The source cannot tell the transformer and the $3\,\Omega$ load from a single $300\,\Omega$ resistor, and $300 = 10^2 \times 3$.

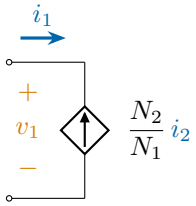
The last line is general. With $v_1 = \frac{N_1}{N_2}v_2$, $i_1 = \frac{N_2}{N_1}i_L$ and $v_2 = R_L i_L$,

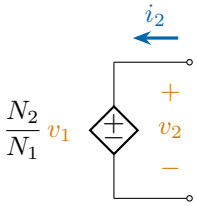
$$\frac{v_1}{i_1} = \left(\frac{N_1}{N_2}\right)^2 R_L.$$

A transformer *rescales resistance* by the square of its turns ratio. That is how an $8\,\Omega$ loudspeaker is made to look like a much larger resistance to the amplifier that drives it.

## 8. A transformer is two dependent sources

The two equations of the ideal transformer can be read as two dependent sources. The secondary is a VCVS controlled by $v_1$; the primary is a CCCS controlled by $i_2$.




The CCCS makes $i_1 = -\frac{N_2}{N_1}i_2$, which is $N_1 i_1 + N_2 i_2 = 0$. The VCVS makes $v_2 = \frac{N_2}{N_1}v_1$. Their powers are $-\frac{N_2}{N_1}v_1 i_2$ and $+\frac{N_2}{N_1}v_1 i_2$: they cancel.

This links the two halves of the lecture. Each dependent source on its own can deliver power; in this pair one always delivers exactly what the other absorbs.

## Summary

| Element | Terminal equations | Power |
|---|---|---|
| VCVS, CCVS | $v = \mu v_1$, $v = r i_1$ | can deliver (active) |
| VCCS, CCCS | $i = g v_1$, $i = \beta i_1$ | can deliver (active) |
| ideal transformer | $v_j/N_j$ all equal, $\sum_j N_j i_j = 0$ | $p = 0$ at every instant |

- A **dependent source** (diamond) takes its value from a voltage or current in another branch. Four kinds: VCVS, CCVS, VCCS, CCCS; the constant's unit is output over input.
- Its terminal equation names a **foreign variable**. Otherwise nothing changes: write KCL, KVL and terminal equations, plus one equation for the controlling variable.
- It can **deliver power** — that is what amplification is — but it **cannot start anything**: without an independent source everything is zero. It is never switched off.
- The **ideal transformer**: $v_1/N_1 = v_2/N_2$ and $N_1 i_1 + N_2 i_2 = 0$, with signs set by the **dots**. It is resistive (no $d/dt$), but a real one needs changing waveforms.
- It **absorbs no power at any instant**, and it rescales a load resistance by $(N_1/N_2)^2$.

### Next lecture

From circuit to **graph**: labelling the nodes and branches of a circuit so that its $2b$ equations can be written systematically — the first step towards node and mesh analysis.

### Reading

- Alexander & Sadiku, *Fundamentals of Electric Circuits*, Section 1.6 (dependent sources) and Section 13.5 (ideal transformers).
- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits* — controlled sources and the ideal transformer.
- Lecture notes and videos of Prof. Emre Tuna, `users.metu.edu.tr/etuna/ee201` — notes p. 11; Lecture 5 video: dependent sources 41:47, the four kinds 43:04–48:14, example 49:15, ideal transformer 51:31, its power 54:47.